Övning 13 - Töjning i 3D

Samband mellan spänning och töjning - Hookes lag

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu(\sigma_y + \sigma_z)) \quad [FS\ 3.1]$$

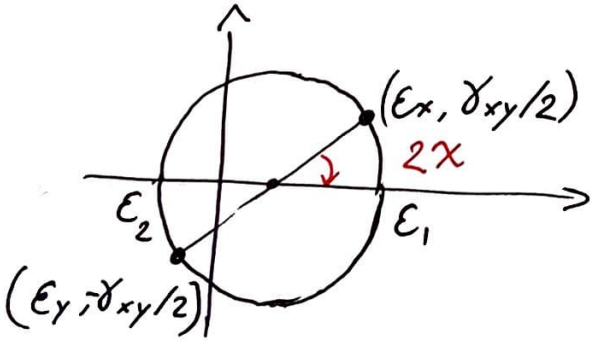
ν - Samband mellan drag och kontraktion, Poissons tal

Rotation av töjningsriktning

$$\varepsilon(\varphi) = \varepsilon_x \cos^2\varphi + \varepsilon_y \sin^2\varphi + \gamma_{xy} \sin\varphi\cos\varphi$$

$$\gamma_{xy}(\varphi) = (\varepsilon_y - \varepsilon_x)\sin(2\varphi) + \gamma_{xy}\cos(2\varphi)$$

Mohrs töjningscirkel



$$\varepsilon_1 = \frac{1}{2}(\varepsilon_x + \varepsilon_y) + R$$

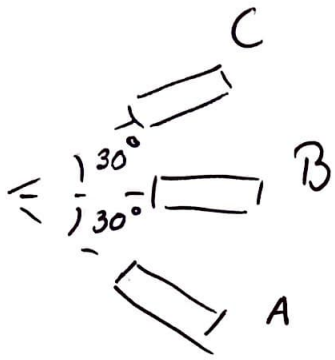
$$\varepsilon_2 = \frac{1}{2}(\varepsilon_x + \varepsilon_y) - R$$

$$R = \left(\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2\right)^{1/2}$$

$$\sin(2\chi) = \frac{\gamma_{xy}}{2R}$$

[FS 2.23]

1.2.10



$\varepsilon_A = 0.0004$

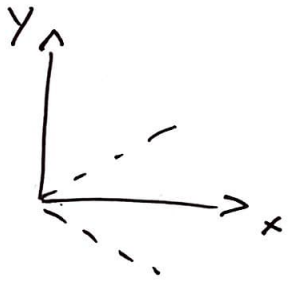
$\varepsilon_B = 0.0006$

$\varepsilon_C = 0.0003$

Sökt: Huvudtöjningar och riktningar

① Ta reda på en töjningsmatris

ex. xy planet $\left(\varepsilon_x, \varepsilon_y, \varepsilon_{xy} = \frac{\gamma_{xy}}{2}\right)$



$\varepsilon_C = \varepsilon(\varphi = 30)$

$\varepsilon_B = \varepsilon_x$

$\varepsilon_A = \varepsilon(\varphi = -30)$

3 obekanta.
3 ekvationer

$$\varepsilon_C = \varepsilon_x \cos^2 30 + \varepsilon_y \sin^2 30 + \gamma_{xy} \sin 30 \cos 30$$

$$\varepsilon_B = \varepsilon_x$$

$$\varepsilon_A = \varepsilon_x \cos^2(-30) + \varepsilon_y \sin^2(-30) + \gamma_{xy} \sin(-30)\cos(-30)$$

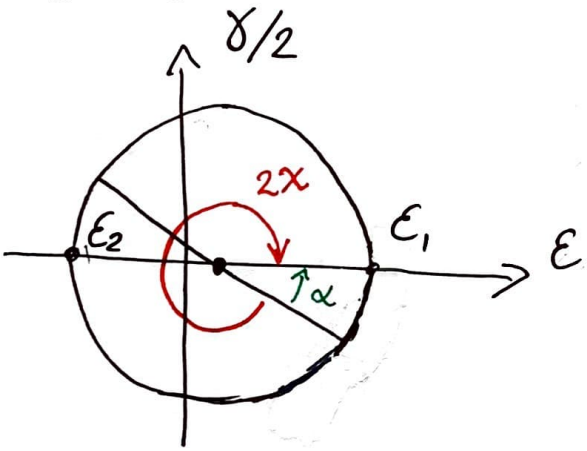
$$\Rightarrow \begin{cases} \varepsilon_x = 6 \cdot 10^{-4} \\ \varepsilon_y = -4 \cdot 10^{-4} \\ \gamma_{xy} = -1.15 \cdot 10^{-4} \Rightarrow \varepsilon_{xy} = -0.58 \cdot 10^{-4} \end{cases}$$

1.2.10 del 2

Töjnings matrisen

$$\underset{\sim}{T} = \begin{bmatrix} 6 & -0.58 & 0 \\ -0.58 & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot 10^{-4}$$

Bestäm huvudtöjningar från mohrs töjningscirkel



$$R = \left(\left(\frac{6 \cdot 10^{-4} - (-4 \cdot 10^{-4})}{2} \right)^2 + (-0.58 \cdot 10^{-4})^2 \right)^{\frac{1}{2}}$$

$$R = 5.01 \cdot 10^{-4}$$

$$\varepsilon_1 = \frac{1}{2}(6 \cdot 10^{-4} + (-4 \cdot 10^{-4})) + 5.01 \cdot 10^{-4}$$

$$\varepsilon_1 = 6.03 \cdot 10^{-4}$$

$$\varepsilon_2 = \frac{1}{2}(6 \cdot 10^{-4} + (-4 \cdot 10^{-4})) - 5.01 \cdot 10^{-4}$$

$$\varepsilon_2 = -4.03 \cdot 10^{-4}$$

Bestäm riktningar från Mohrs töjningscirkel

$$2\chi = 360 - \alpha$$

$$\chi = \frac{360 - 6.59}{2}$$

$$\boxed{\chi_1 = 176.3^\circ}$$

Alt $\chi_1 = -3.7^\circ$

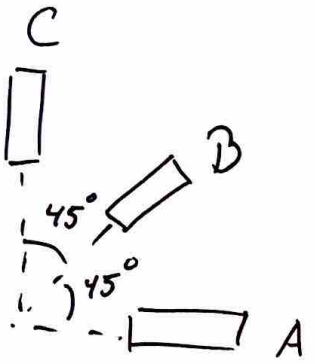
$$\alpha = \operatorname{asin}\left(\frac{\gamma_{xy}/2}{R} \right)$$

$$\alpha = 6.59^\circ$$

$$\boxed{\chi_2 = \chi_1 + 90 = 266.3^\circ}$$

$\chi_2 = 86.3^\circ$

1.2.12



$\varepsilon_A = 0.12$

$\varepsilon_B = 0.02$

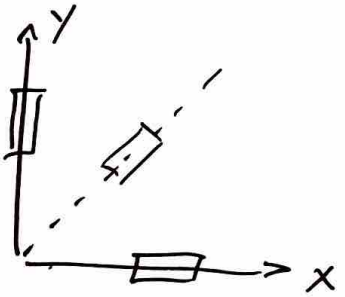
$\varepsilon_C = -0.08$

Sökt: Finns det någon riktning där töjningen är 0?

$\varepsilon(\varphi) = 0$ för vilken φ?

$$\varepsilon(\varphi) = \varepsilon_x \cos^2\varphi + \varepsilon_y \sin^2\varphi + \gamma_{xy} \sin\varphi\cos\varphi$$

Bestäm $\varepsilon_x, \varepsilon_y, \gamma_{xy}$



$\varepsilon_x = \varepsilon(0) = \varepsilon_A$

$\varepsilon_y = \varepsilon(90) = \varepsilon_C$

$\varepsilon(45) = \varepsilon_B$

$$\varepsilon_B = \varepsilon(45) = \varepsilon_A \cos^2 45 + \varepsilon_C \sin^2 45 + \gamma_{xy} \sin 45 \cos 45$$

$$\Rightarrow \quad \gamma_{xy} = 2\varepsilon_B - \varepsilon_A - \varepsilon_C = 0$$

$$\underset{\sim}{T} = \begin{bmatrix} 0.12 & 0 \\ 0 & -0.08 \end{bmatrix}$$

1.2.12

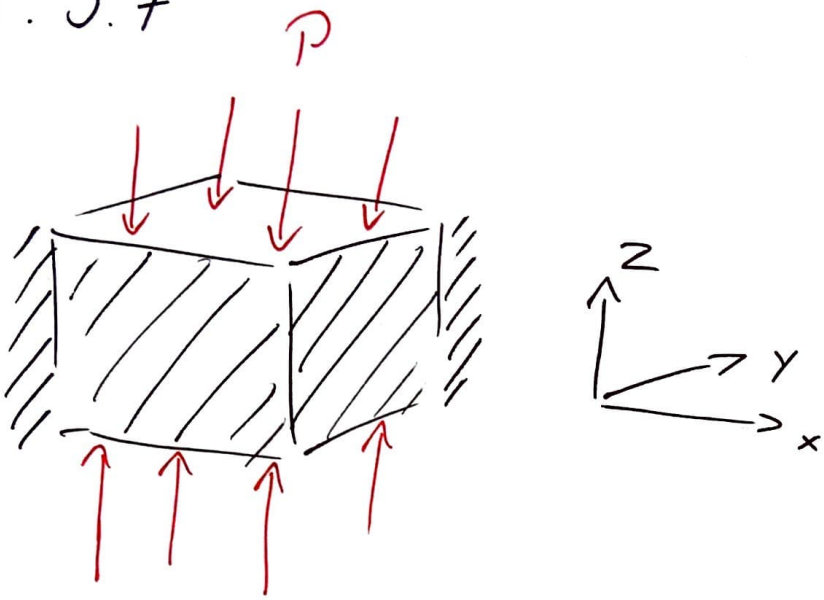
Bestäm vinkel då $\mathcal{E}(\varphi)=0$

$$\mathcal{E}(\varphi) = 0.12\cos^2\varphi - 0.08\sin^2\varphi = 0$$

$$\Rightarrow \quad \tan^2\varphi = \frac{0.12}{0.08}$$

$$\Rightarrow \quad \boxed{\varphi = 50.8^\circ} \quad \text{och} \quad \boxed{\varphi = -50.8^\circ} \quad \boxed{\varphi = 129.2^\circ}$$

1.3.7



Givet: E, ν

Sökt: Trycket kuben avger på väggarna. σ_x och σ_y

Vi vet: $\sigma_z = -P$ $\quad \varepsilon_x = \varepsilon_y = 0$

Bestäm σ_x och σ_y från Hookes lag i 3D

$$\sigma_x = \frac{E}{1+\nu}\left(\varepsilon_x + \frac{\nu}{1-2\nu}(\varepsilon_x + \varepsilon_y + \varepsilon_z)\right)$$

$$\sigma_x = \frac{E}{1+\nu}\left(\frac{\nu}{1-2\nu}\varepsilon_z\right)$$ ε_z - okänd

$\sigma_y = \sigma_x$ på grund av symmetri

$$\sigma_z = \frac{E}{1+\nu}\left(\varepsilon_z + \frac{\nu}{1-2\nu}(\varepsilon_x + \varepsilon_y + \varepsilon_z)\right) =$$

$$\sigma_z = \frac{E}{1+\nu}\left(\varepsilon_z + \frac{\nu}{1-2\nu}\varepsilon_z\right) = -P$$

1.3.7 del 2

$$\Rightarrow \quad \varepsilon_z \frac{E}{1+\nu}\left(\frac{1-\nu}{1-2\nu}\right) = -P$$

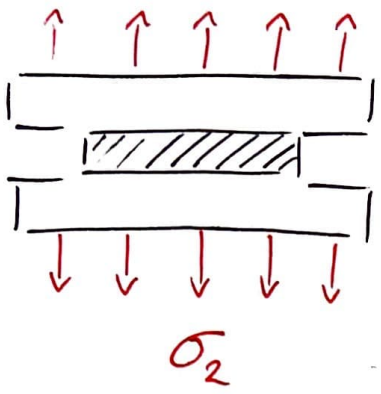
$$\boxed{\varepsilon_z = -\frac{P(1+\nu)(1-2\nu)}{(1-\nu)E}}$$

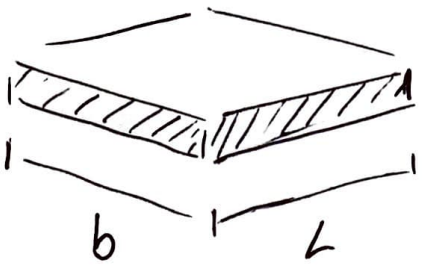
Sätt in ε_z i spänningarna σ_x, σ_y

$$\sigma_x = \sigma_y = \frac{E}{1+\nu}\left(\frac{\nu}{1-2\nu}\right)(-P)\left(\frac{1+\nu}{E}\right)\left(\frac{1-2\nu}{1-\nu}\right)$$

$$\boxed{\sigma_x = \sigma_y = -P\frac{\nu}{1-\nu}}$$

1. 3.10





$h << b, L$

Sökt: a) Beräkna $E' = \frac{\sigma_z}{\varepsilon_z}$ b) Visa $E' >> E$ om inkompressibelt mat.

Vi vet: $\varepsilon_x = \varepsilon_y = 0$ (Ingen glidning)

① Bestäm σ_z från Hookes lag

$$\sigma_z = \frac{E}{1+\nu}\left(\varepsilon_z + \frac{\nu}{1-2\nu}(\varepsilon_x + \varepsilon_y + \varepsilon_z)\right)$$

$$\sigma_z = \frac{E}{1+\nu}\,\frac{(1-\nu)}{1-2\nu}\,\varepsilon_z$$

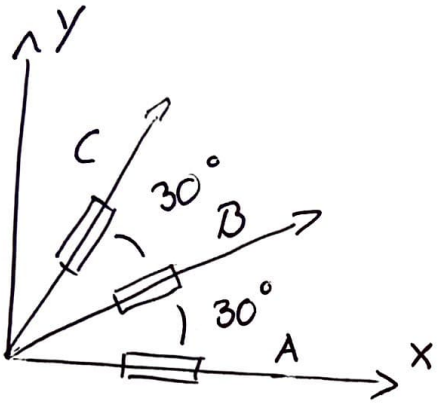
② Sätt in σ_z i ekv. E'

$$E' = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}$$

③ Inkompressibilitet innebär $\nu = 0.\overline{5}$

$$E'(\nu = 0.\overline{5}) = \frac{E(1-0.\overline{5})}{(1+0.\overline{5})(1-1)} = \infty$$

1.3.14



$$\varepsilon_A = 1.4 \cdot 10^{-4}$$

$$\varepsilon_B = 1.2 \cdot 10^{-4}$$

$$\varepsilon_C = -0.2 \cdot 10^{-4}$$

Material: $E = 70\,GPa \quad \nu = 0.35$

Sökt: a) Spänningstillståndet $\underset{\sim}{S}$

b) Huvudspänningar och riktningar

Bestäm $\varepsilon_x, \varepsilon_y, \gamma_{xy}$ för att sätta in i Hookes

$$\varepsilon_A = \varepsilon_x$$

$$\varepsilon_B = \varepsilon(30°) = \varepsilon_x \cos^2 30 + \varepsilon_y \sin^2 30 + \gamma_{xy} \sin 30 \cos 30$$

$$\varepsilon_C = \varepsilon(60°) = \varepsilon_x \cos^2 60 + \varepsilon_y \sin^2 60 + \gamma_{xy} \sin 60 \cos 60$$

$$\Longrightarrow \begin{cases} \varepsilon_x = 1.4 \cdot 10^{-4} \\ \varepsilon_y = -1.4 \cdot 10^{-4} \\ \gamma_{xy} = 1.155 \cdot 10^{-4} \end{cases}$$

Hookes lag

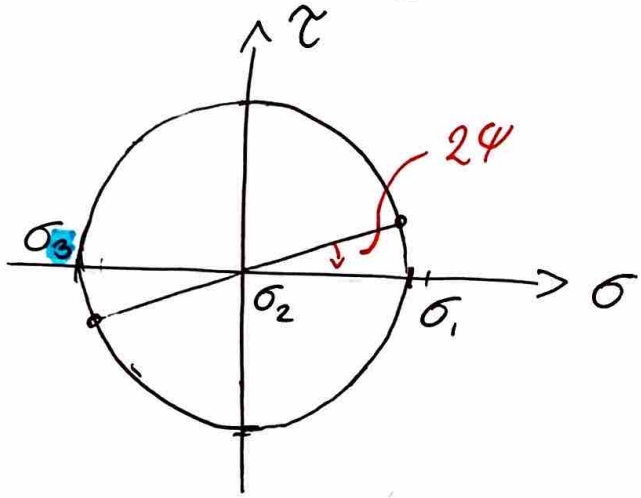
$$\sigma_x = \frac{E}{1-\nu^2}(\varepsilon_x - \nu \varepsilon_y) \qquad \tau_{xy} = \frac{E}{2(1+\nu)} \gamma_{xy}$$

$$\sigma_y = \frac{E}{1-\nu^2}(\varepsilon_y - \nu \varepsilon_x)$$

1.3.14 del 2

$$\Rightarrow \begin{cases} \sigma_x = 7.26\ MPa \\ \sigma_y = -7.26\ MPa \\ \tau_{xy} = 2.99\ MPa \end{cases}$$

Huvudspänningar och riktningar



$$\sin 2\varphi = \frac{\tau_{xy}}{R}$$

$$\sigma_1 = \frac{1}{2}(\sigma_x + \sigma_y) + R$$

$$\sigma_3 = \frac{1}{2}(\sigma_x + \sigma_y) - R$$

$$R = \left(\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2\right)^{\frac{1}{2}}$$

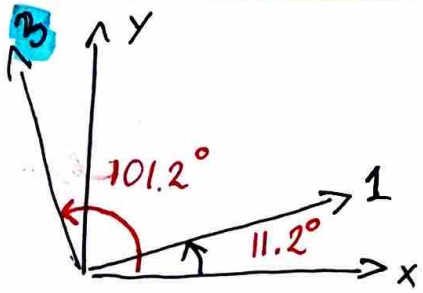
$$R = 7.85\ MPa$$

$$\sigma_1 = 7.85\ MPa$$

$$\sigma_2 = 0\ MPa$$

$$\sigma_3 = -7.85\ MPa$$

$$\sin 2\varphi_1 = \frac{\tau_{xy}}{R} \Rightarrow \varphi_1 = 11.2^\circ \Rightarrow \varphi_3 = \varphi_1 + 90^\circ = 101.2^\circ$$



Rotation fysiskt plan